Vanshil Shah

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Education

University of Pennsylvania — Aug. 2021 - May 2023
M.S.E. Robotics — GPA: 3.91/4
Courses: Machine Learning, Theoretical Deep Learning, Learning in Robotics, Geometric Computer Vision, Deep Learning for Vision, Advanced Robotics, Convex Optimization

Nirma University — Aug 2015 - May 2019
B.E. in Mechanical Engineering — GPA: 8.1/10

Publications

Prashant Kumar*, Sabyasachi Sahoo*, **Vanshil Shah**, Vineetha Kondameedi, Abhinav Jain, Akshaj Verma, Chiranjib Bhattacharyya, Vinay V. **"DSLR : Dynamic to Static LiDAR scan Reconstruction using adversarially trained autoencoder"**(***Proceedings of the AAAI Conference on Artificial Intelligence 2021)***

Unni Krishnan R Nair*, Anish Gupta*, D. A. Sasi Kiran, Ajay Shrihari, **Vanshil Shah**, K. Madhava Krishna **"Non Holonomic Collision Avoidance under Non-Parametric Uncertainty: A Hilbert Space Approach"**(***Proceedings of European Control Conference(ECC), 2021***)

Work Experience

NVIDIA, San Jose, CA — Feb 2024 - Ongoing
Deep Learning Algorithm Engineer

- Shipped **10+ production inference microservices** for large language and vision-language models to enterprise customers, implementing custom backend changes in vLLM and TensorRT-LLM.
- Implementd TRT-LLM encoder-decoder for a document-parsing VLM (Nemoretriever-Parse), achieving **3x throughput** over the baseline; shipped as part of NV Ingest blueprint at NVIDIA GTC.
- Built end-to-end encoder-decoder workflow tests and CI-based performance regression detection across multiple inference backends.

Quidient, Columbia, MD — July 2023 - Jan 2024
Computer vision engineer, 3D Reconstruction team — Brevin Tilmon

- Developed core Generalized Scene Reconstruction (GSR) algorithms for Quidient Reality®, producing relightable, millimeter-accurate 5D models from smartphone video, outperforming NeRF and Gaussian Splatting baselines.
- Developed ***CUDA-Enabled*** light physics module which increased the speed of reconstruction by **6 times**.
- Adopted Nsight Compute for profiling, ensuring streamlined kernel operations and effective memory oversight.

Ford Motors, Autonomous Vehicles LLC, Palo Alto — May 2022 - August 2022
Localisation and Mapping intern, Perception Team — Dr. Punarjay Chakravarthy

- Trained Neural Radiance Fields aiding synthetic data generation for downstream CV tasks.
- Developed a slot attention-based neural radiance field for disentangling background and foreground.
- Achieved comparable performance of image reconstruction metrics like ***PSNR, LPIPS and SSIM*** on both real world and simulated dataset. ***[Project Report] [Slides]***

Indian Institute of Science(IISc), Bangalore — Nov 2019 - Sept 2020
Software engineer, Machine Learning Lab | Collaboration: Ati Motors — Sabyasachi Sahoo

- Integrated Google Cartographer SLAM algorithm with our ***Generative adversarial model***.
- Achieved ***4 times better reconstruction*** on Chamfer Distance over state of the art baselines.

Projects

Segmentation and Object Detection

- **SOLO:** Implemented the network proposed in paper: *Segmenting Objects by Location* to predict instance segmentation masks over 3 categories(Vehicle, People and Animals) on COCO dataset *Github*
- **Faster RCNN:** Implemented a 2-stage RCNN based object classifier. This involved training the first stage Region Proposal Network and second stage regressor, and classifier. MAP achieved: 0.76 *Github*

Neural Image Caption Generation with Visual Attention

- Trained an LSTM network with soft attention mechanism to generate natural language captions for images *Github*

Geometric computer vision

- **Multi view stereo reconstruction**: Implementation of two-view stereo and multi-view stereo algorithms for dense 3d reconstruction.Implemented the 2 view SFM algorithm using SIFT features and 8-point algorithm in tandem with RANSAC for robust camera pose estimation *Github*

Localization and Estimation

- **Particle filter based SLAM:** Integrated the inertial orientation and odometry with a 2D LIDAR scan to build the occupancy grid map of the environment while localising the robot using a particle filter *Github*
- **Orientation tracking with inertial data:** Implemented a Quaternion based Unscented Kalman Filter(UKF) to track 3D orientation from Gyroscope and Accelerometer data *Github*

Technical Skills

Languages: Python, C/C++, MATLAB **Software Tools**: CUDA, Git, Docker, CMake
Libraries: PyTorch, Sklearn, NumPy, pandas, Matplotlib, OpenCV